# Healthcare Machine Learning Project
# Summary Report

## 1. Introduction

This project applies supervised machine learning to three real-world healthcare datasets. The goal was to go through the complete machine learning workflow from loading and cleaning data to training models, tuning hyperparameters, evaluating results, and comparing findings to published research. I worked with the Pima Indians Diabetes dataset, the Cleveland Heart Disease dataset, and the Medical Insurance Costs dataset. Two of those are binary classification problems, and the third is a regression problem, which lets me practice both types of evaluation metrics. All models were built using scikit-learn in Python, and hyperparameter tuning was done using GridSearchCV with 5-fold cross-validation throughout.

## 2. Datasets Overview

### 2.1 Pima Indians Diabetes (Classification)

This dataset comes from the National Institute of Diabetes and Digestive and Kidney Diseases and has 768 records from adult Pima Indian women aged 21 and older. There are eight input features, including glucose level, BMI, blood pressure, insulin, age, number of pregnancies, skin thickness, and a diabetes pedigree function. The target variable is binary, where 1 means diabetic and 0 means not diabetic. About 34.9% of the records are positive cases, making this a moderately imbalanced dataset. The main preprocessing challenge was that several features used zero as a placeholder for missing measurements. Glucose, blood pressure, BMI, insulin, and skin thickness cannot realistically be zero in a living person, so I replaced those zeros with the median of the non-zero values for each column. I chose this dataset because it is one of the most cited datasets in medical ML and has plenty of published benchmarks to compare against.

### 2.2 Cleveland Heart Disease (Classification)

This dataset was originally collected at the Cleveland Clinic Foundation and made available through the UCI Machine Learning Repository. It contains 297 patient records with 13 clinical features, including age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, ST depression, slope, number of major vessels, and a thal value. The binary target called condition indicates whether heart disease was detected. The class split is fairly balanced at 160 negative and 137 positive cases. All features are already numeric, so preprocessing was straightforward, and I just applied StandardScaler for models that need it. I chose this dataset because the mix of clinical features is interesting, and it has a strong academic citation history.

### 2.3 Medical Insurance Costs (Regression)

This dataset has 1,338 insurance beneficiary records with six features: age, sex, BMI, number of children, smoker status, and region. The target variable is charges, which is the annual medical cost billed by health insurance in US dollars. This is the only regression task in the project. Preprocessing involved label encoding for sex and smoker, and one-hot encoding for region. The dataset has no missing values at all, which made it quick to prepare. I chose it specifically to complement the two classification datasets and give the project coverage of regression metrics like MAE, RMSE, and R-squared.

## 3. Models and Approach

### 3.1 Diabetes - Logistic Regression, Random Forest, MLP

For the diabetes dataset, I trained three models. Logistic Regression was my baseline because it is simple, interpretable, and gives a good starting point for a binary classification problem. Random Forest was my second choice because it handles nonlinear interactions between features well and does not require scaling. My third model was a Multi-Layer Perceptron neural network, which can learn more complex patterns at the cost of being harder to interpret. Logistic Regression and MLP used scaled features from StandardScaler, while Random Forest used the raw features.

### 3.2 Heart disease - KNN, Decision Tree, Naive Bayes

I chose a different set of three algorithms for the heart disease dataset to demonstrate a broader range of methods. KNN classifies new points based on the majority label of their nearest neighbors and is sensitive to scale, so I used the scaled version of the data for it. Decision Tree learns interpretable if-then rules and does not need scaling. Naive Bayes is a probabilistic classifier that assumes feature independence given the class, an assumption that often holds well enough on clinical data where features are not too strongly correlated.

### 3.3 Insurance - Linear Regression, Decision Tree, Random Forest

For the regression task, I used Linear Regression as the baseline, then a Decision Tree Regressor to capture nonlinear effects, and finally a Random Forest Regressor as the ensemble method. The smoker variable causes charges to spike dramatically in a way that is hard for a linear model to handle, which is exactly why I included the tree-based models here.

## 4. Results

### 4.1 Pima Diabetes - Classification Results (After Tuning)

After hyperparameter tuning with GridSearchCV, all three models performed well within the expected range for this dataset. Random Forest was the top performer across all five metrics.

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.773 | 0.698 | 0.620 | 0.657 | 0.828 |
| Random Forest | 0.799 | 0.745 | 0.648 | 0.693 | 0.851 |
| MLP Neural Network | 0.786 | 0.729 | 0.630 | 0.676 | 0.839 |

Glucose and BMI were the two most important features according to the Random Forest feature importance scores, which aligns with what clinicians would expect for diabetes prediction.

### 4.2 Heart Disease - Classification Results (After Tuning)

Naive Bayes was the best performer on the heart disease dataset, which makes sense given that many of the clinical features in this dataset are relatively independent of each other.

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| KNN | 0.797 | 0.800 | 0.763 | 0.781 | 0.864 |
| Decision Tree | 0.780 | 0.771 | 0.763 | 0.767 | 0.780 |
| Naive Bayes | 0.814 | 0.818 | 0.763 | 0.790 | 0.876 |

### 4.3 Insurance Costs - Regression Results (After Tuning)

Random Forest was again the best model with an R-squared of 0.863. Smoker status was by far the most important feature, followed by age and BMI.

| Model | MAE | RMSE | R2 |
|---|---|---|---|
| Linear Regression | $2,641 | $4,181 | 0.771 |
| Decision Tree | $2,895 | $4,730 | 0.738 |
| Random Forest | $2,427 | $3,745 | 0.863 |

## 5. Comparison to Prior Work

For the Pima Diabetes dataset, I referenced a survey by Kavakiotis et al. (2017) published in the Computational and Structural Biotechnology Journal. That paper reviewed 85 ML studies on diabetes data and found that Random Forest and SVM consistently achieved accuracies between 77 and 79 percent. My Random Forest hit 79.9 percent, which is at the top of that range. Addressing the zero-imputation preprocessing step that many prior studies missed likely helped push results slightly higher.

For the Heart Disease dataset, the original data was introduced by Detrano et al. (1989) in the American Journal of Cardiology. More recent Kaggle benchmarks put Random Forest at 80 to 85 percent accuracy and KNN at 75 to 82 percent on this dataset. My KNN scored 79.7 percent, and my Naive Bayes scored 81.4 percent, both within those published ranges.

For the insurance regression task, Rashid and Abdulrahman (2022) in the International Journal of Online and Biomedical Engineering reported Linear Regression at R-squared of 0.74 to 0.78 and Random Forest at 0.85 to 0.87. My Linear Regression scored 0.771, and my Random Forest scored 0.863, closely matching their findings. Both studies identified smoker status as the dominant predictor, which validates the result across two independent implementations.

## 6. Discussion and Reflection

The most consistent finding across all three datasets was that Random Forest outperformed the other models in every case. Ensemble methods reduce variance by averaging many trees, which helps especially when individual features are noisy or correlated. For future healthcare ML projects, I would start with Random Forest or gradient boosting as the baseline rather than a linear model.

There are a few honest limitations to this project. The Pima dataset only covers Pima Indian women aged 21 and older, so a model trained on it would not generalize well to other populations without retraining. The Heart Disease dataset has only 297 rows, which means performance estimates have meaningful variance. The insurance dataset is clean and useful for a class project, but it is missing real-world features like medical history and pre-existing conditions that would matter in an actual actuarial application.

Overall, this project was genuinely useful for understanding the full ML pipeline from end to end. Preprocessing decisions, especially the zero-imputation step on the diabetes dataset, had a more noticeable effect on results than I expected. Hyperparameter tuning consistently improved performance over baseline models, though often by a smaller margin than I anticipated, which suggests that good preprocessing and model selection matter more than fine-tuning in many practical cases.

## 7. References


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**[4]** Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.